ratchet release lever is pressed at the top of the hand brake, the pawl is moved up and releases the ratchet. However while pulling the lever in upward direction, cable is also pulled towards the upward direction. The cable actuates the mechanical brake which is connected to it and stops the rear wheel by applying force. Moreover ratchet release lever releases and moves the pawl towards downward direction with a help of spring and again connects the ratchet. By this way hand brake is operated to stop the vehicle.

6.8 Common parts of braking system

The common parts of brake system are listed below

1. Brake drum
2. Brake shoes
3. Brake linings

6.8.1 Brake drum

Brake drums are commonly opened in the inner side and closed on the outer side. The brake shoes present inside the brake drum are expanded and operate the brake. Due to the friction created while operating the brake system the brake drum is heated up and the temperature of the drum is increased. To reduce this temperature, fins are located at outside the brake drum. The brake drum should not wear due to friction caused inside while rotating and it must have high thermal conductivity to dissipate heat quickly. Moreover it must be of less weight and it must have enough strength. Brake drums are mostly made up of cast iron or nickel chromium steel alloy or Aluminium alloy materials.

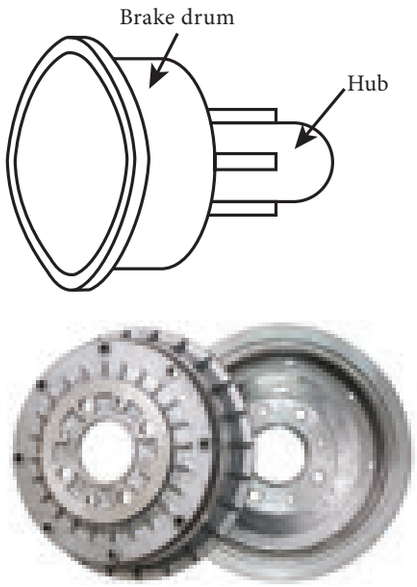


Figure 6.8.1 Brake drum

6.8.2 Brake shoes

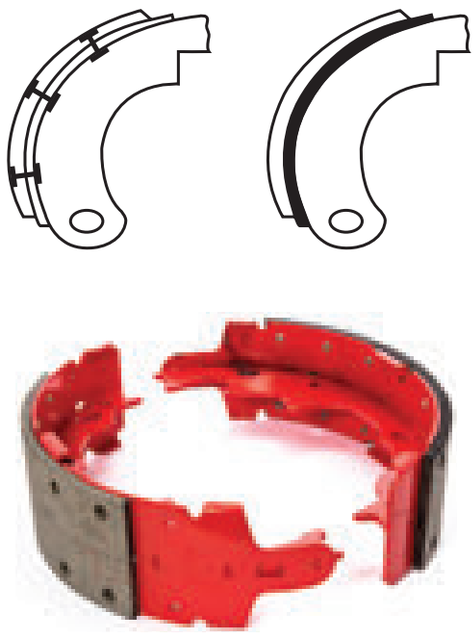


Figure 6.8.2 Brake shoes

In each brake carrier plate two brake shoes are provided. The brake shoes looks like in "T" shaped cross section. The right hand side of the brake shoe is called as the primary brake shoe and the shoe in the left hand side is is called as the secondary brake shoe. The bottom edges of the two brake shoes are connected

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